



SWARNANDHRA COLLEGE OF ENGINEERING AND TECHNOLOGY (AUTONOMOUS)

Department of CSE – Cyber Security

Evaluation of Machine Learning Algorithms for the Detection of Fake Bank Currency

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ABSTRACT

- Fake bank currency circulation is a major economic and security concern.
- Manual detection of counterfeit notes is difficult and unreliable.
- Currency images are processed and classified as real or fake using ML algorithms.
- The system aims to improve accuracy, efficiency, and reliability in currency authentication.

INTRODUCTION

- Fake currency circulation is a serious economic and security issue.
- Manual detection of fake notes is unreliable
- Machine Learning helps in automated detection
- Project focuses on detecting fake banknotes



EXISTING SYSTEM

- Detection is performed using traditional and rule-based approaches.
- KNN and similar models are used for basic classification.
- Entire dataset is considered during prediction.
- Existing systems rely on limited feature similarity methods.
- Performance varies based on train-test ratio.

DISADVANTAGES OF EXISTING SYSTEM

- Provides less accuracy for complex counterfeit notes.
- Detection process is low in efficiency.
- Not suitable for real-time large-scale usage.
- Performance degrades with high-dimensional data.
- Limited adaptability to new counterfeit patterns.

PROPOSED SYSTEM

- Multiple supervised ML algorithms are implemented.
- Algorithms include SVM, RF, LR, NB, DT, and KNN.
- Different train-test ratios are used for evaluation.
- Performance measured using quantitative metrics.
- System automatically classifies currency as genuine or fake.

ADVANTAGES OF PROPOSED SYSTEM

- Achieves high detection accuracy.
- Provides high efficiency and faster processing.
- Reduces human intervention and errors.
- Suitable for real-time applications.
- Scalable and adaptable for future datasets.

SRS (Software Requirement Specification)

Hardware Requirements:

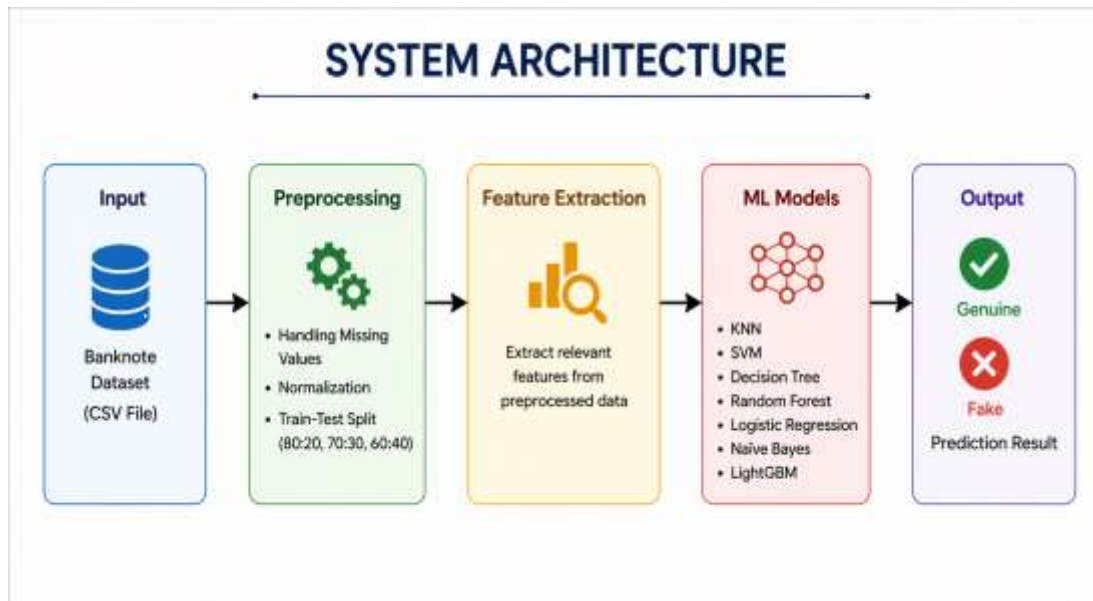
- System with minimum 4 GB RAM
- Intel i3 processor or above
- Minimum 10 GB free storage
- Webcam / Scanner

Software Requirements:

- Windows / Linux
- Python
- OpenCV, Scikit-learn, NumPy, Pandas
- Matplotlib

SYSTEM ARCHITECTURE

- Input currency dataset is collected.
- Data preprocessing is performed.
- Feature extraction is applied.
- Machine learning models are trained and tested.
- Output classifies currency as genuine or fake.



MODULES

- User Interface Module
- Dataset Upload Module
- Data Preprocessing Module
- Machine Learning Training Module
- Model Evaluation Module
- Prediction Module

User Interface Module

- Provides interaction between user and system
- Allows dataset upload
- Displays prediction result (Genuine / Fake)
- Shows model accuracy

Dataset Upload Module

- Accepts UCI banknote dataset
- Validates CSV file format
- Checks data integrity
- Stores data for processing

Data Preprocessing Module

- Handles missing values
- Normalizes features
- Splits dataset (80:20, 70:30, 60:40)
- Converts data for ML training

Machine Learning Model Training Module

Algorithms Used:

- KNN
- SVM
- Decision Tree
- Random Forest
- Logistic Regression
- Naïve Bayes
- LightGBM

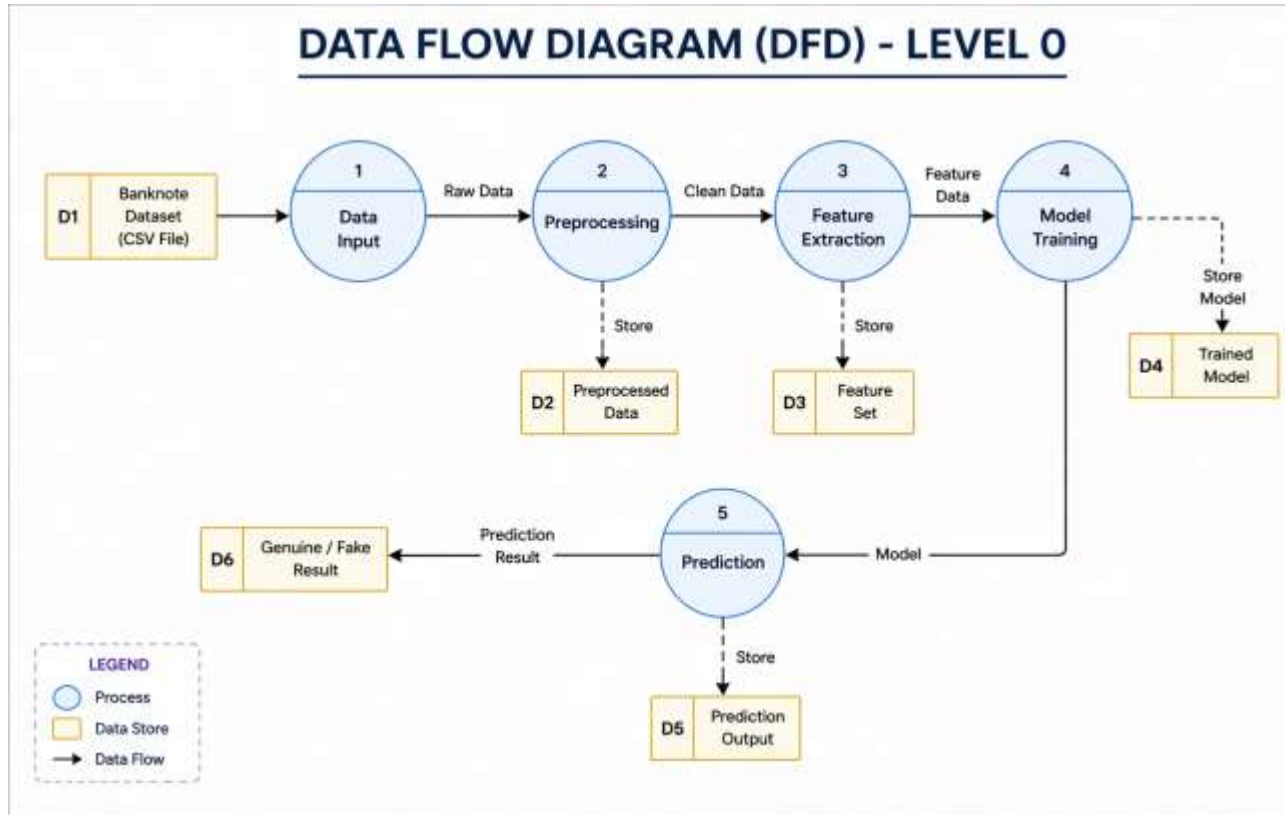
Model Evaluation & Comparison

- Accuracy
- Precision
- Recall
- F1-Score

Prediction Module

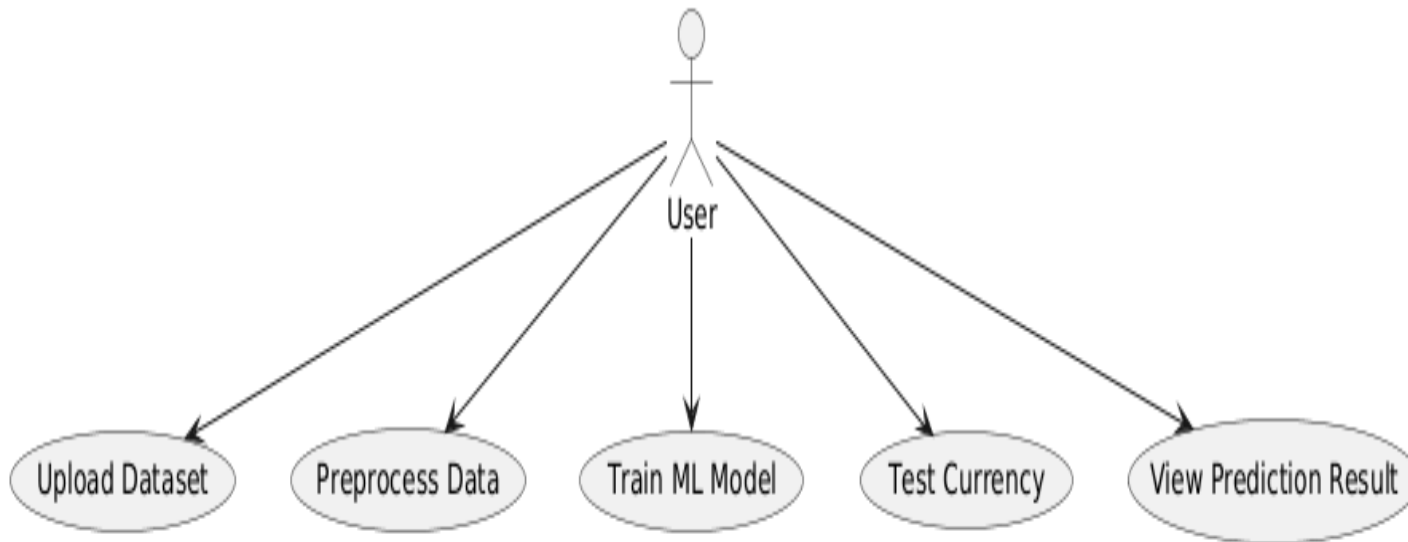
- Accepts new currency data
- Applies best trained model
- Outputs: Genuine / Fake

Data Flow Diagram



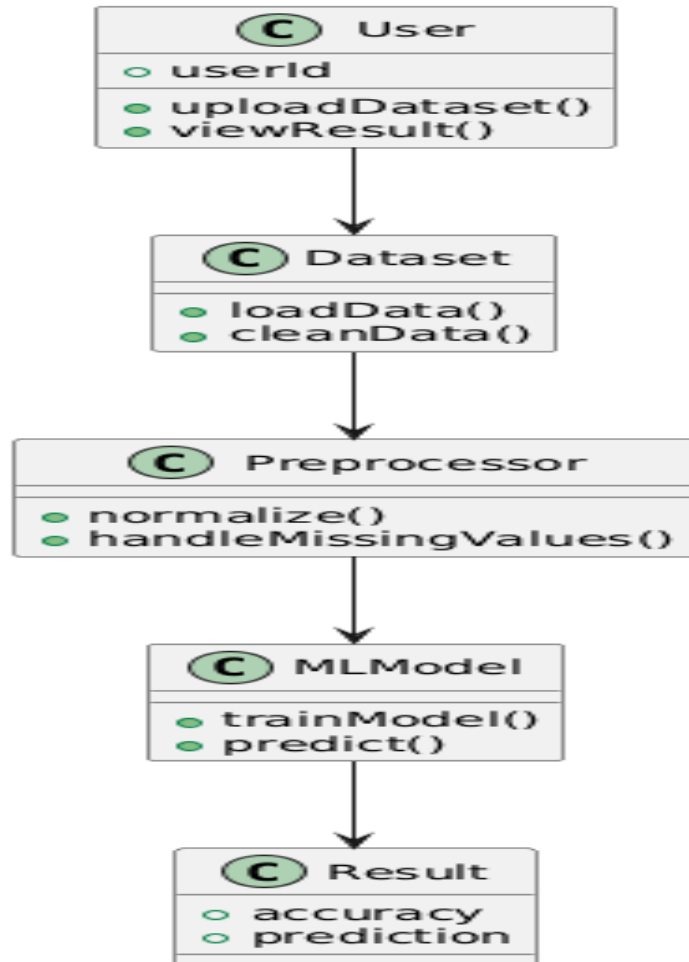
UML Diagrams

Use Case Diagram



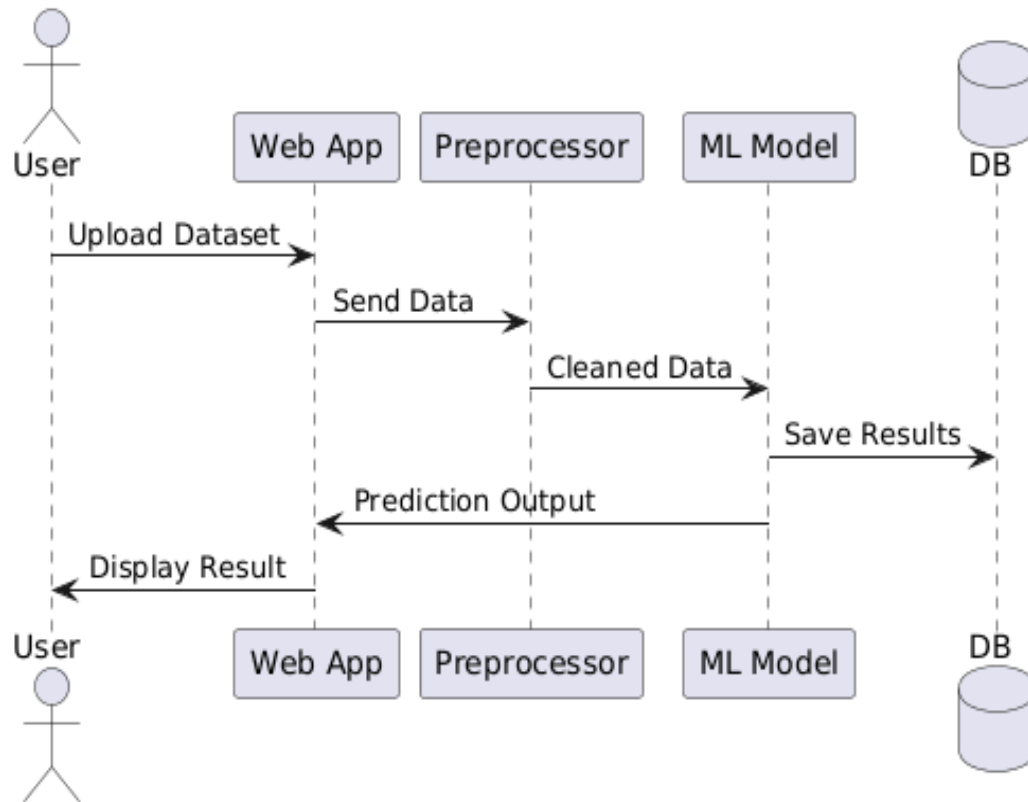
UML Diagrams

Class Diagram



UML Diagrams

Sequence Diagram - Prediction Flow



Methodology (Techniques)

- Data Collection
- Data Preprocessing
- Feature Extraction
- Model Training
- **Model Evaluation Techniques:**
 - Accuracy
 - Precision
 - Recall
 - F1 Score

Algorithms Used

- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- Logistic Regression
- Naïve Bayes
- **LightGBM:** A fast gradient boosting algorithm that uses decision trees for high accuracy and efficient fake currency detection.

Output Screenshots

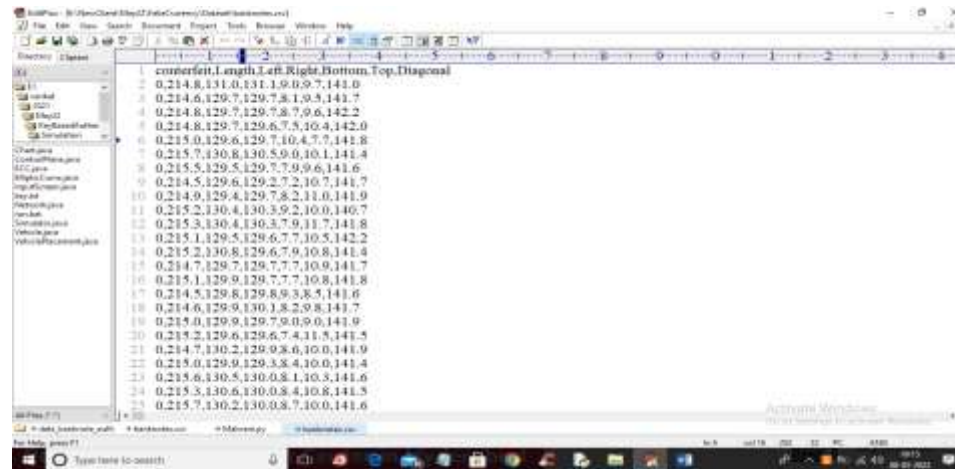


Fig no.1 Dataset Screen



Fig no.2 Main Interface of Fake Currency Detection System

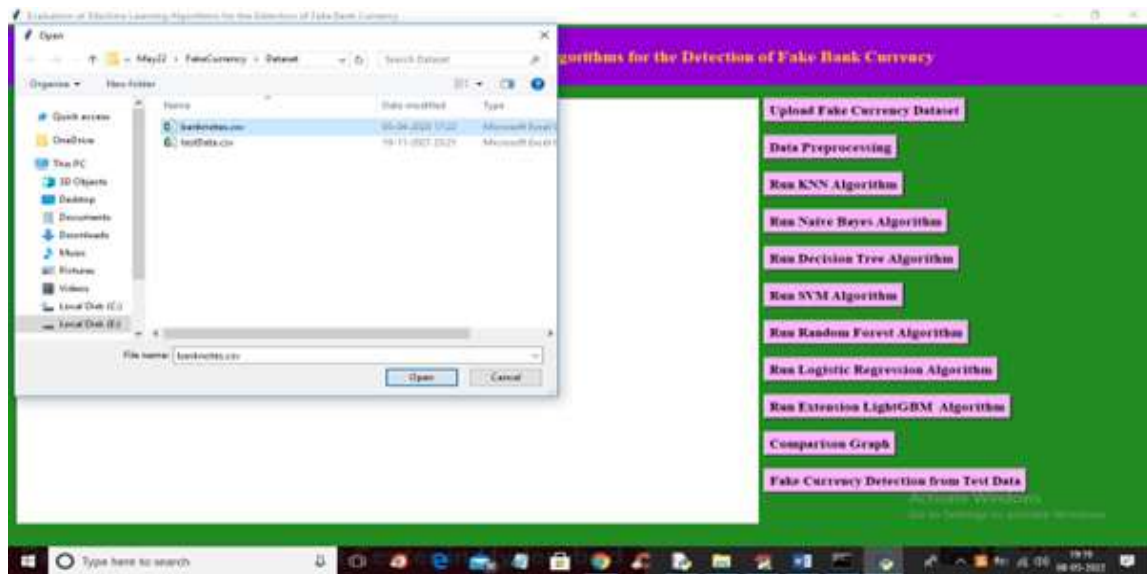


Fig no.3 Uploading Dataset into the System

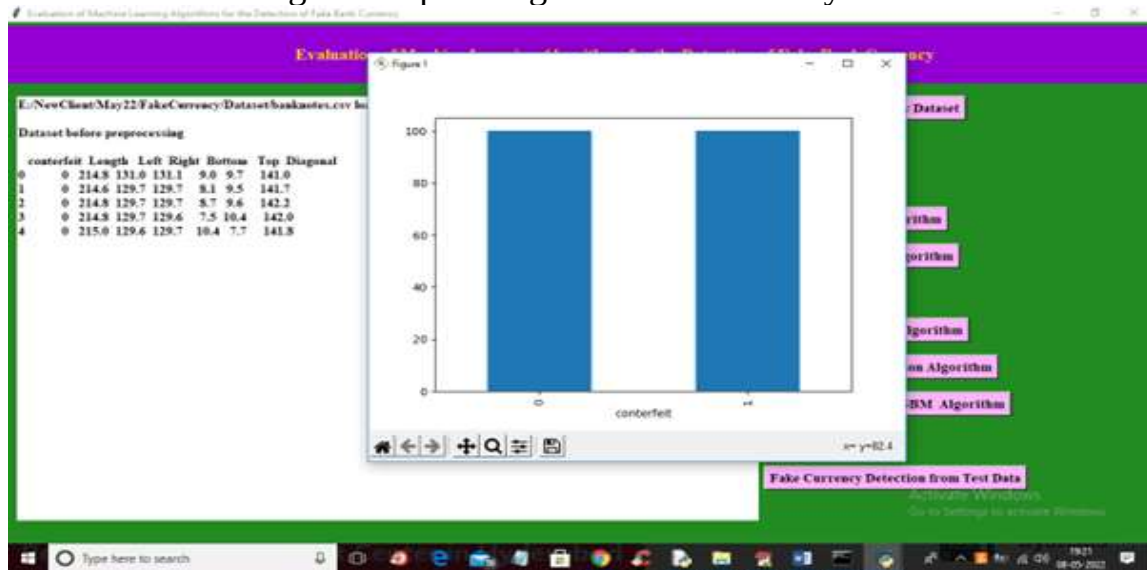


Fig no.4 Dataset Preprocessing Output

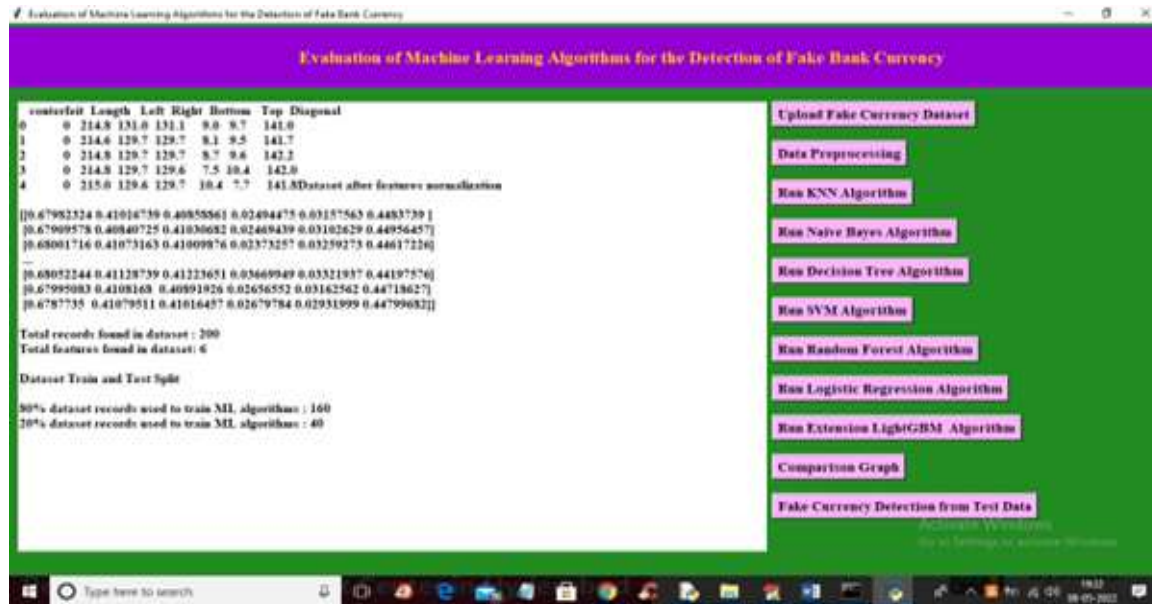


Fig no.5 Dataset Normalization and Train-Test Split



Fig no.6 KNN Algorithm Accuracy Output



Fig no.7 Accuracy Results of All Algorithms



Fig no.8 LightGBM Algorithm Output and Comparison

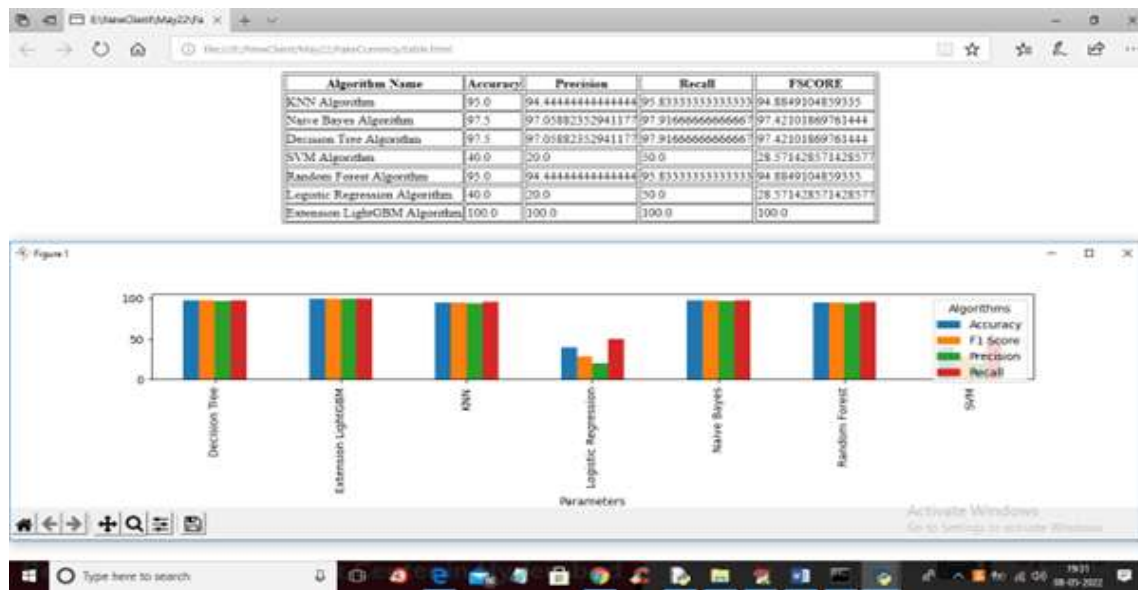


Fig no.9 Comparison Graph of Algorithm Performance

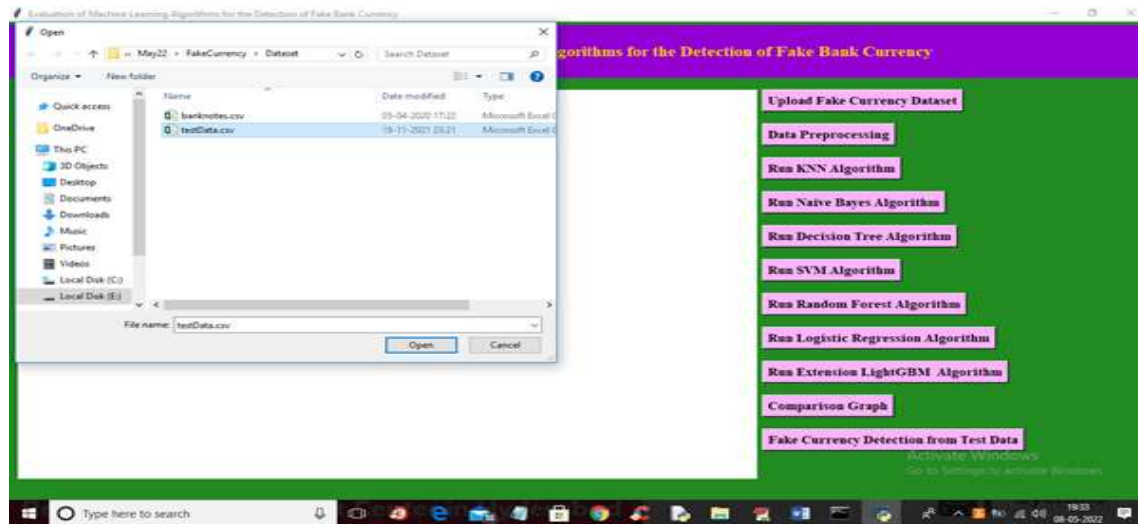


Fig no.10 Uploading Test Data for Prediction



Fig no.11 Fake Currency Detection Output

Conclusion

This project uses machine learning to detect fake currency accurately and efficiently. By comparing different algorithms, the best model is selected for reliable prediction. The system reduces manual effort and improves security, making it useful for real-world applications.

Future Scope

The system can be improved using deep learning techniques like CNN for better accuracy. It can be developed into a mobile application and integrated with ATMs for real-time currency detection, as well as extended to support multiple currencies.

References

- UCI Machine Learning Dataset
- Research Papers on Fake Currency Detection
- IEEE / Journal Papers

THANK YOU